



CENG 241 OBJECT ORIENTED PROGRAMMING

Car Rental System

Team Members

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1. INTRODUCTION

The Car Rental System is designed to enhance the experience for both businesses and customers by automating and streamlining the traditional car rental process.

1.1 Purpose

The car rental system aims to simplify car rental management systems, addressing complexity issues, and ensuring data privacy. It seeks to offer administrators a systematic view of data. The program is designed to create a sample car rental system and provide a user-friendly system for administrators and customers.

Simultaneously, it assists businesses in managing their operations more effectively, leading to improved efficiency and cost reduction.

1.2 Scope

In the project scope, we gathered and analyzed data for a car rental system, creating a system plan. This plan was translated into code using an algorithm, and the application's functionality was illustrated through a class diagram. Rigorous testing was conducted to ensure the program's proper functioning.

2. DESIGN

2.1 Approach

The program is designed to meet the needs of administrators and customers involved in managing car rental operations, with a priority on simplicity and fast access to essential information.

2.2 System

The main page of the program is as in the figure.

If they are staff, they can press 1 and enter to the staff menu. If they are consumer, they can press 2 and enter to the customer menu.

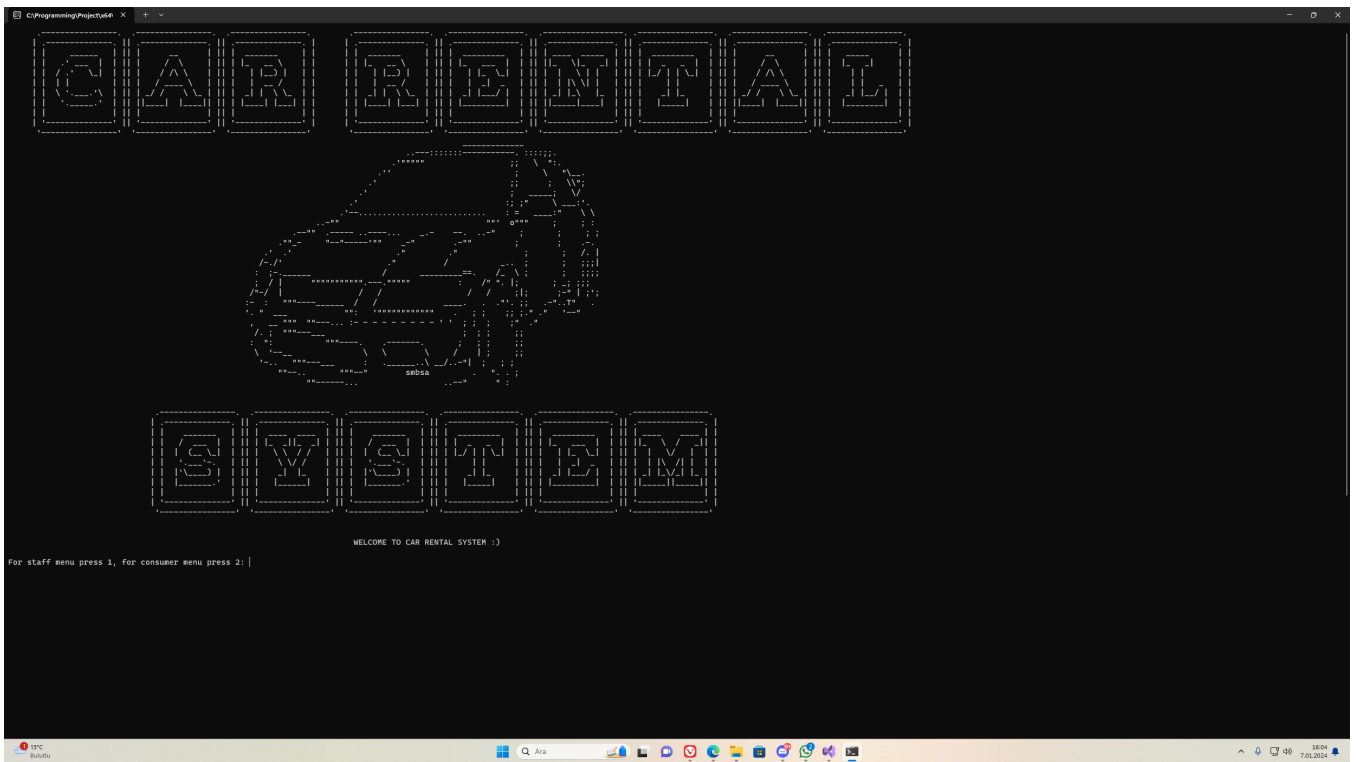


Figure 1. Main Page

With this program;

Staff can enter the system by using their username and password. Can add new cars, can delete cars, can see the statistics and can log out. Customers can list all cars, can list all cities and locations and can rent cars if they are available then log out.

We run the program by entering the number of the operation we want in the select section.

For example, we can enter as a staff using our id and password.

We have to enter our username and password correctly to enter the system.

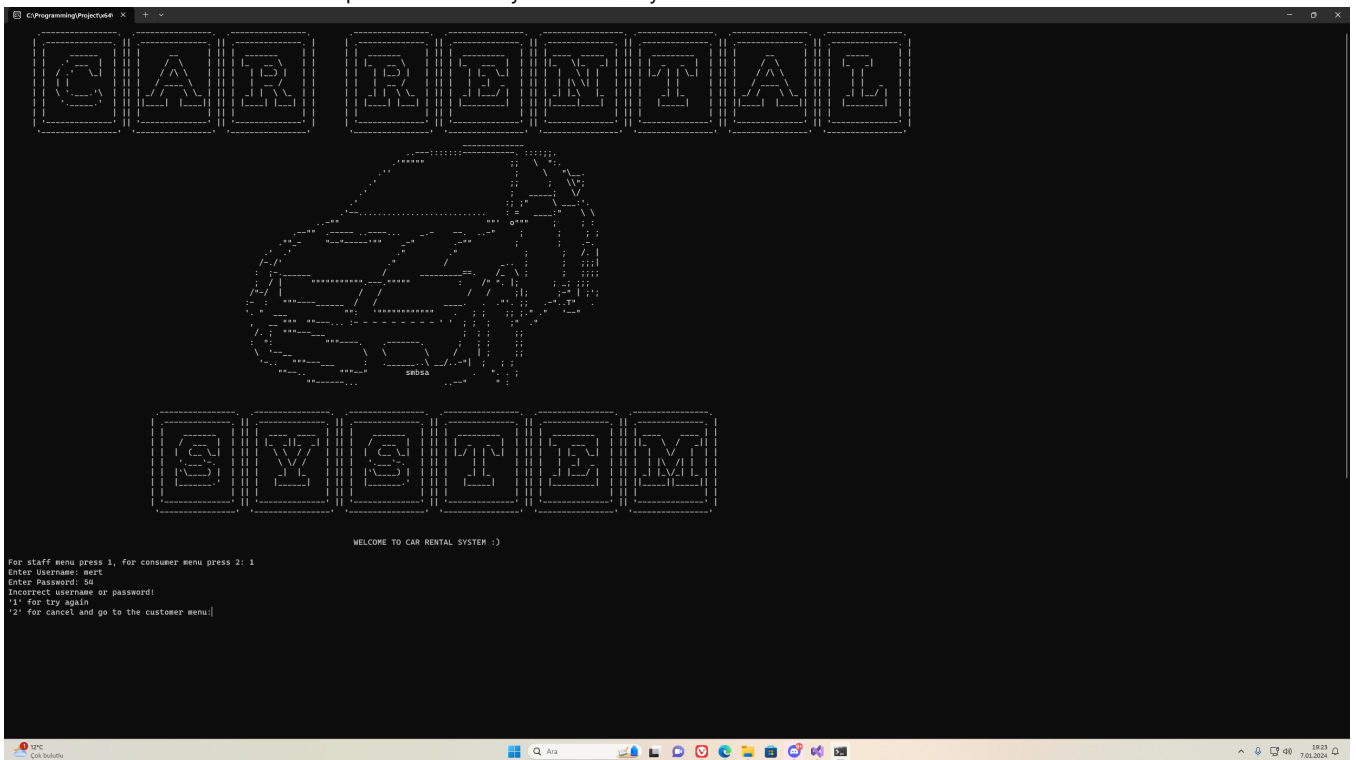


Figure 2. Staff Login

Since we logged in, we continue our process by choosing the first operation.

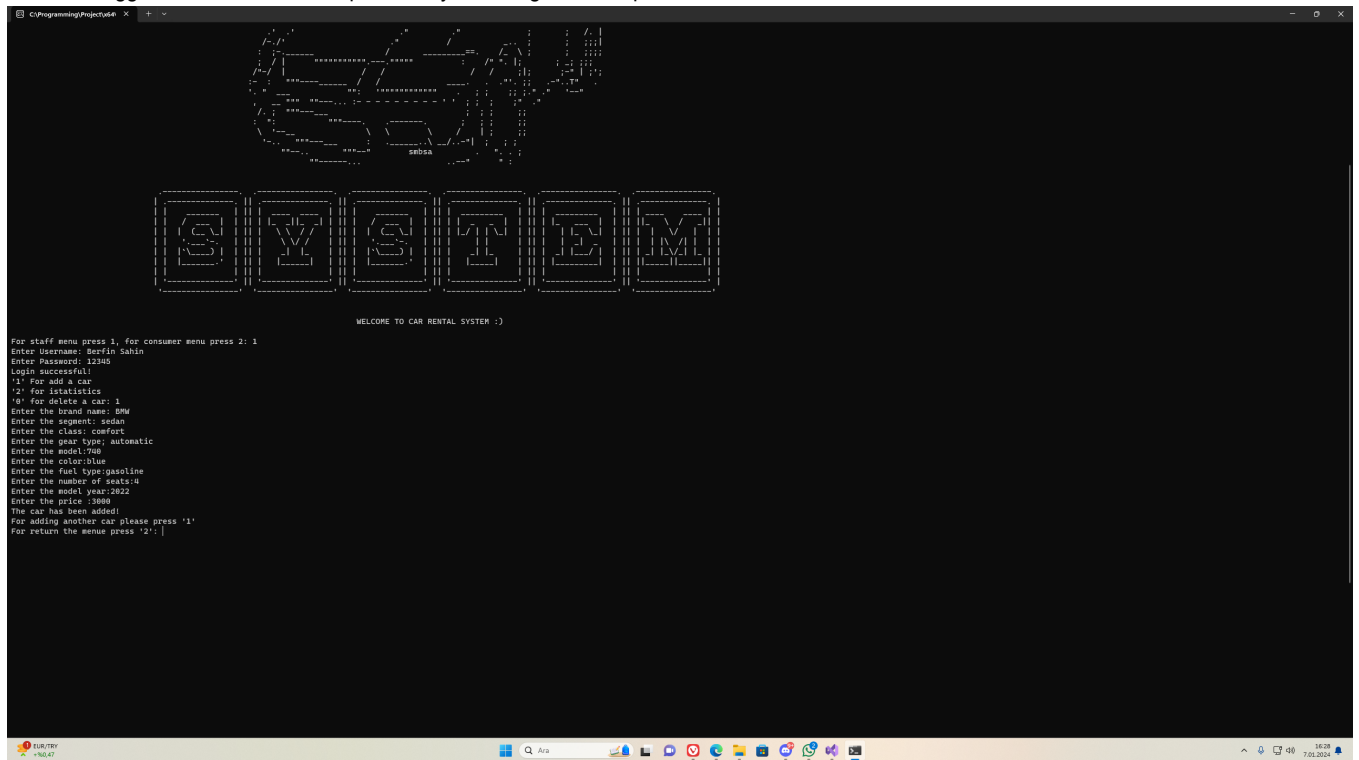


Figure 3. Staff, Operation 1

We added a new car to our database.

We can continue our process by choosing the third operation.

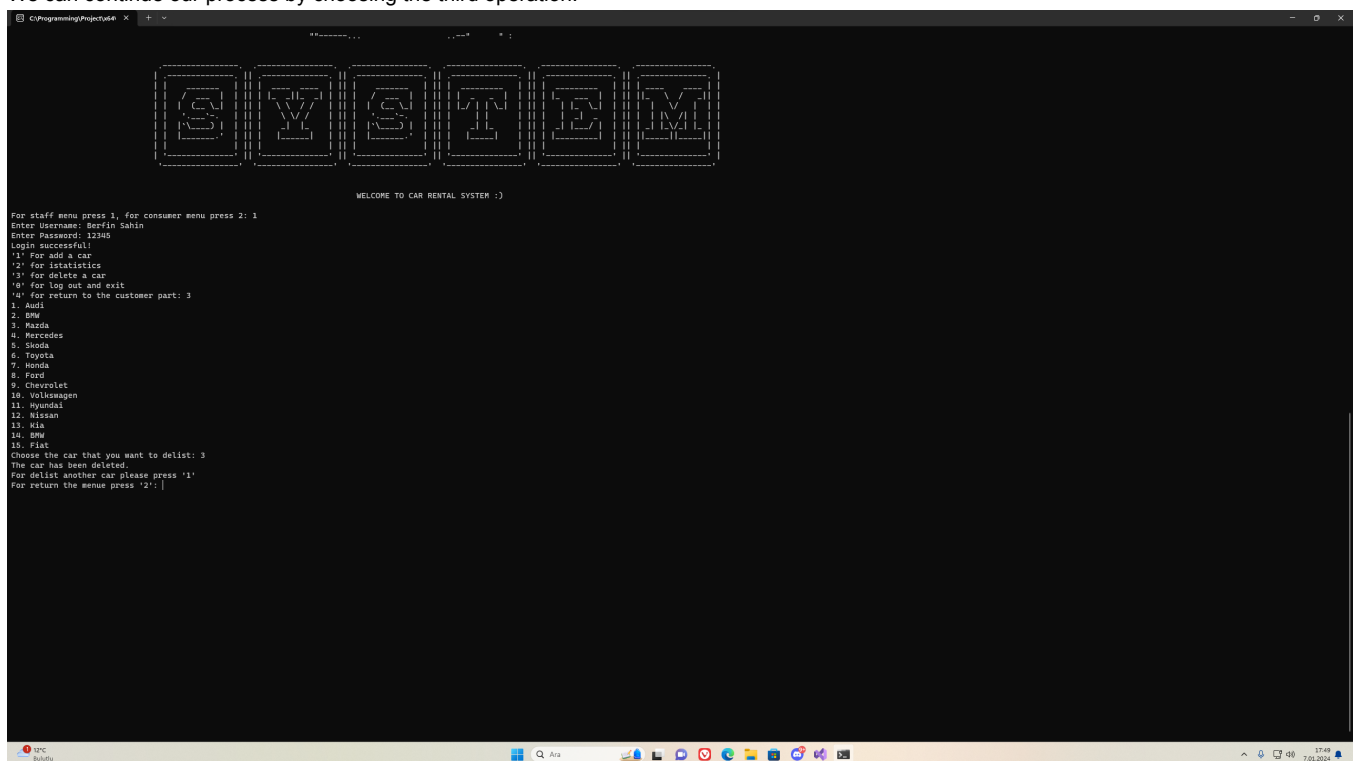


Figure 4. Staff, Operation 3

We deleted a car from our system.

We can enter as a customer and see the customer menu. Then continue our process by choosing the first operation.

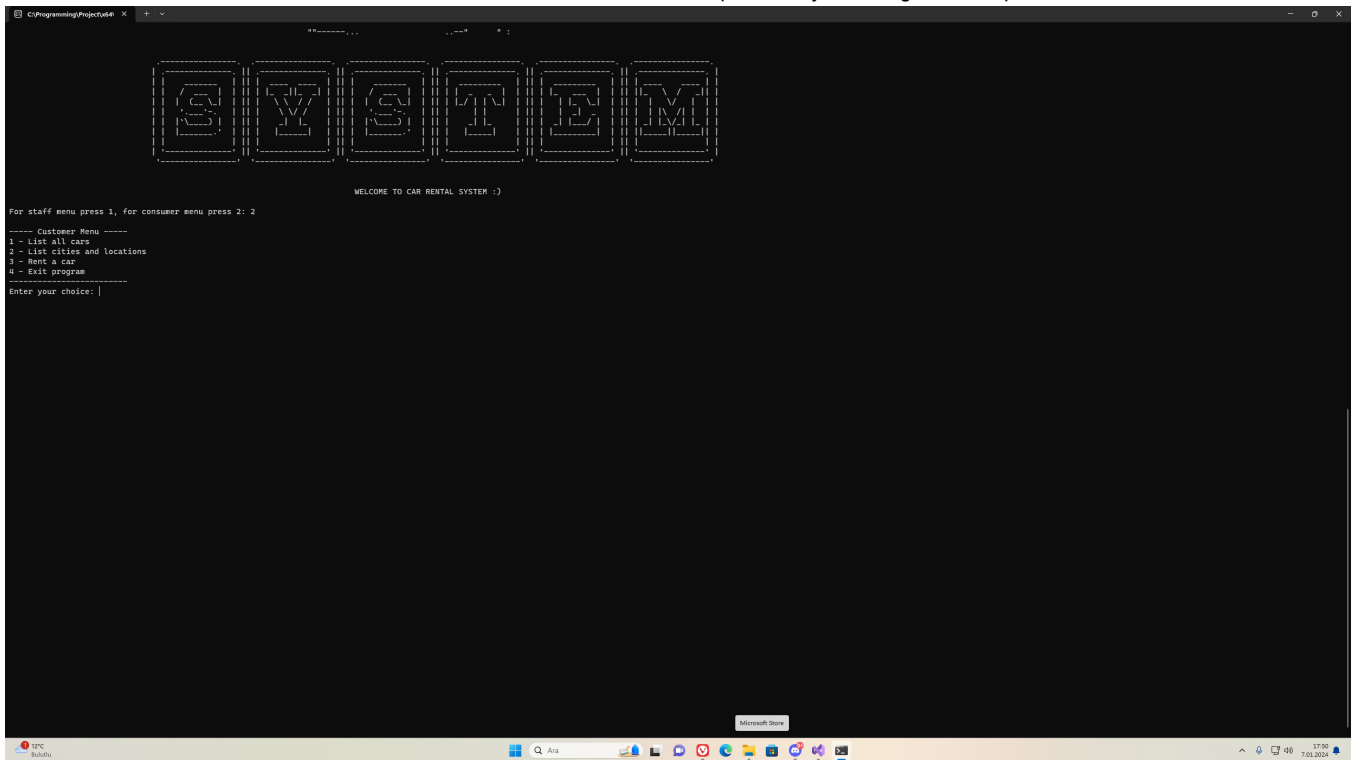


Figure 5. Customer Menu

We can continue our process by choosing the first operation. We can list all the cars this way.

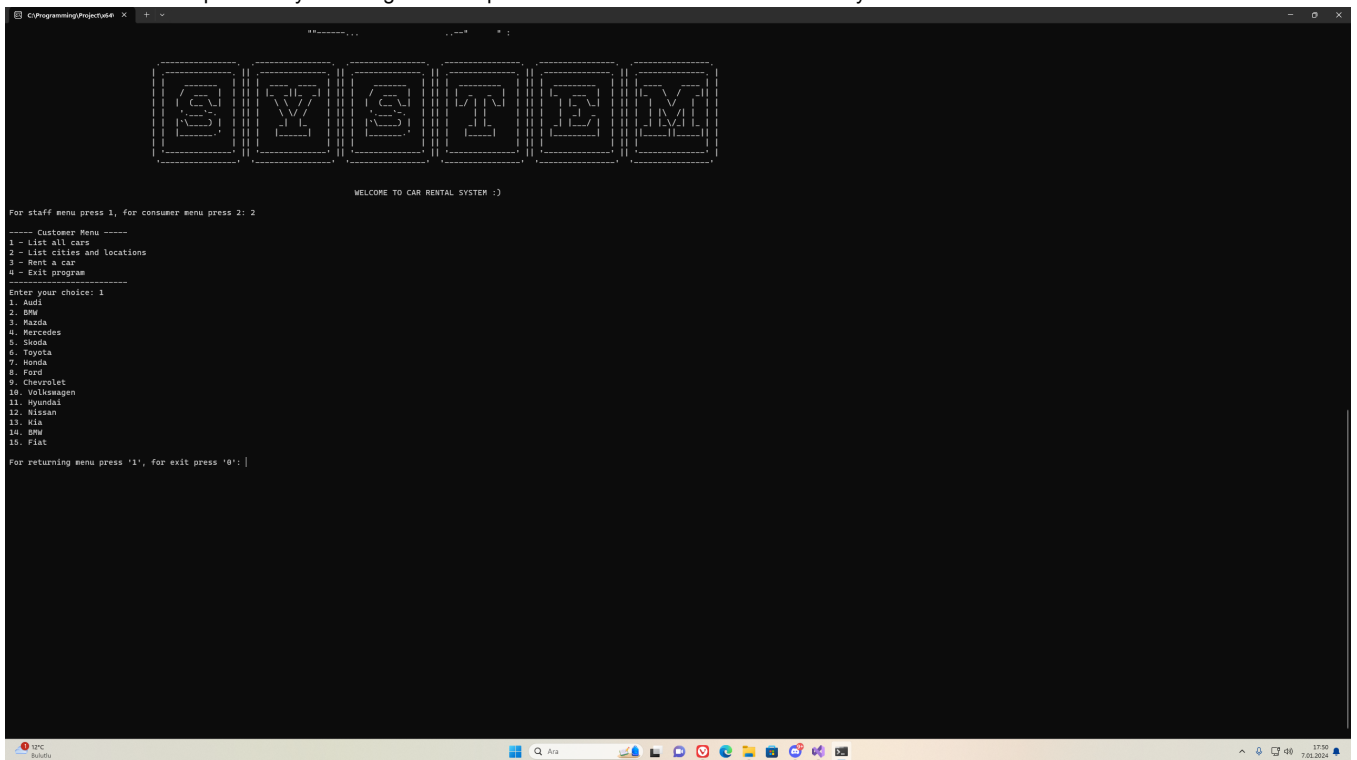


Figure 6. Customer, Operation 1

We can continue our process by choosing the second operation.

```
C:\Programming\Project\ide x + -
----- Customer Menu -----
1 - List all cars
2 - List cities and locations
3 - Rent a car
4 - Exit program
-----
Enter your choice: 1
1. Audi
2. BMW
3. Mazda
4. Mercedes
5. Honda
6. Toyota
7. Nissan
8. Ford
9. Chevrolet
10. Volkswagen
11. Hyundai
12. Nissan
13. Kia
14. BMW
15. Fiat

For returning menu press '1', for exit press '0': 1

----- Customer Menu -----
1 - List all cars
2 - List cities and locations
3 - Rent a car
4 - Exit program
-----
Enter your choice: 2
1. Ankara
Konumlar:
-> Esenboga Havalimani
-> Atli Otogar
-> Ankara Gari

2. Istanbul
Konumlar:
-> Sabiha Gokcen Havalimani
-> Istanbul Havalimani
-> Taksim

3. Samsun
Konumlar:
-> Carsamba Havalimani
-> Cumhuriyet Meydani

4. Izmir
Konumlar:
-> Adnan Menderes Havalimani
-> Izmir Otogari

5. Antalya
Konumlar:
-> Antalya Havalimani
-> Antalya Otogari

6. Bursa
Konumlar:
-> Yenisehir Havalimani

7. Mugla
Konumlar:
-> Milas-Bodrum Havalimani

For returning menu press '1', for exit press '0': |
```

Figure 7. Customer, Operation 2

We listed all the cities and locations.

We can continue our process by choosing the third operation. We can rent a car.

```
C:\Programming\Project\ide x + -
3. Samsun
Konumlar:
-> Carsamba Havalimani
-> Cumhuriyet Meydani

4. Izmir
Konumlar:
-> Adnan Menderes Havalimani
-> Izmir Otogari

5. Antalya
Konumlar:
-> Antalya Havalimani
-> Antalya Otogari

6. Bursa
Konumlar:
-> Yenisehir Havalimani

7. Mugla
Konumlar:
-> Milas-Bodrum Havalimani

For returning menu press '1', for exit press '0': 1

----- Customer Menu -----
1 - List all cars
2 - List cities and locations
3 - Rent a car
4 - Exit program
-----
Enter your choice: 3
Start date (yyyy-mm-dd): 2020-01-01
End date (yyyy-mm-dd): 2020-01-10
Cities We Serve:
1. Ankara
2. Istanbul
3. Samsun
4. Izmir
5. Antalya
6. Bursa
7. Mugla
Please select a city:
4
Konumlar:
-> Esenboga Havalimani
-> Atli Otogar
-> Ankara Gari

Available Cars:
1. Audi A3
2. BMW M3
3. Mazda Miata
10. Fiat Egea
Please choose a car (1-10): 2
The reservation has been completed successfully.
Price that you have to pay for this car: 5000000
Please enter your name and surname: Muhammed Selimcan Bicer
Please enter your credit card number: 2546857950123508
Please enter your credit card CVC: 254
Please enter your credit card expiration date(yyy/dd): 28/01

Thank you for choosing us, you can take your car from one of our service points.

Customer name: Muhammed Selimcan Bicer
Customer credit card number: 2546857950123508
Customer CVC: 254
Customer credit card expiration date: 28/01
Do you want to continue? (yes: 1, No: 0)
for staff entry press '2': |
```

Figure 8. Customer, Operation 3

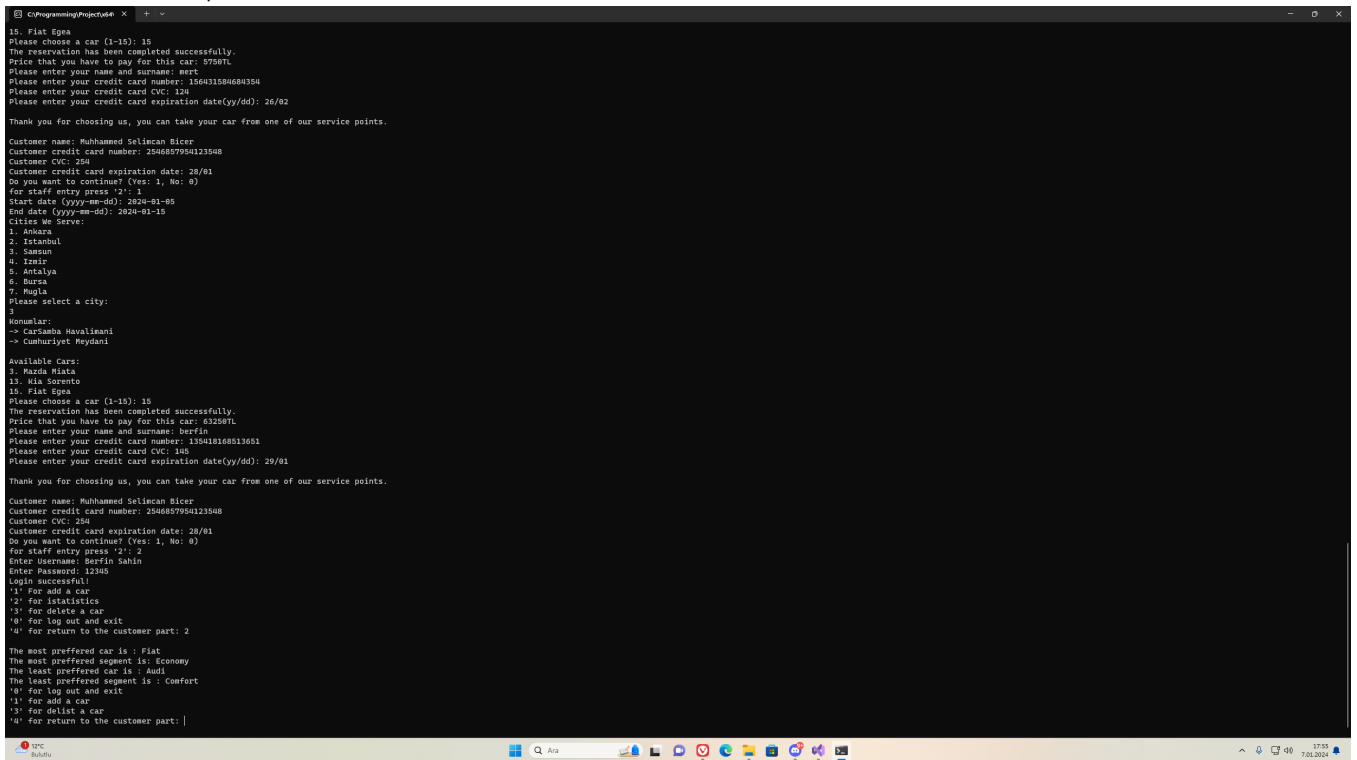
-We added integrated a date-based reservation system into our program. This system displays available cars based on the selected date and location. If a car is already reserved for the date chosen by the customer, the system will not display it as available on that specific date.-

We selected a date range, then selected a location.

System showed us the available cars based on our preferences.

Then we entered our paying information and completed the process.

Since we rented multiple different cars, we can enter as a staff and see the statistics.



```
C:\Programming\Project>
15. Fiat Egea
Please choose a car (1-15): 15
The reservation has been completed successfully.
Price that you have to pay for this car: 5780TL
Please enter your name and surname: mert
Please enter your credit card number: 15643158664354
Please enter your credit card CVC: 124
Please enter your credit card expiration date(yy/dd): 26/02
Thank you for choosing us, you can take your car from one of our service points.

Customer name: Muhammed Selincan Bicer
Customer credit card number: 2566857954123508
Customer CVC: 254
Customer credit card expiration date: 28/01
Do you want to continue? (Yes: 1, No: 0)
For staff entry press '2': 1
Start date (yyyy-mm-dd): 2024-01-05
End date (yyyy-mm-dd): 2024-01-15
Cities We Serve:
1. Ankara
2. Istanbul
3. Samsun
4. Izmir
5. Antalya
6. Bursa
7. Mugla
Please select a city:
3
KonuLar:
-> Carsamba Havalimani
-> Cumhuriyet Meydani

Available Cars:
1. Mazda Miata
13. Kia Sorento
15. Fiat Egea
Please choose a car (1-15): 15
The reservation has been completed successfully.
Price that you have to pay for this car: 63250TL
Please enter your name and surname: Berfin
Please enter your credit card number: 135419168013401
Please enter your credit card CVC: 145
Please enter your credit card expiration date(yy/dd): 29/01
Thank you for choosing us, you can take your car from one of our service points.

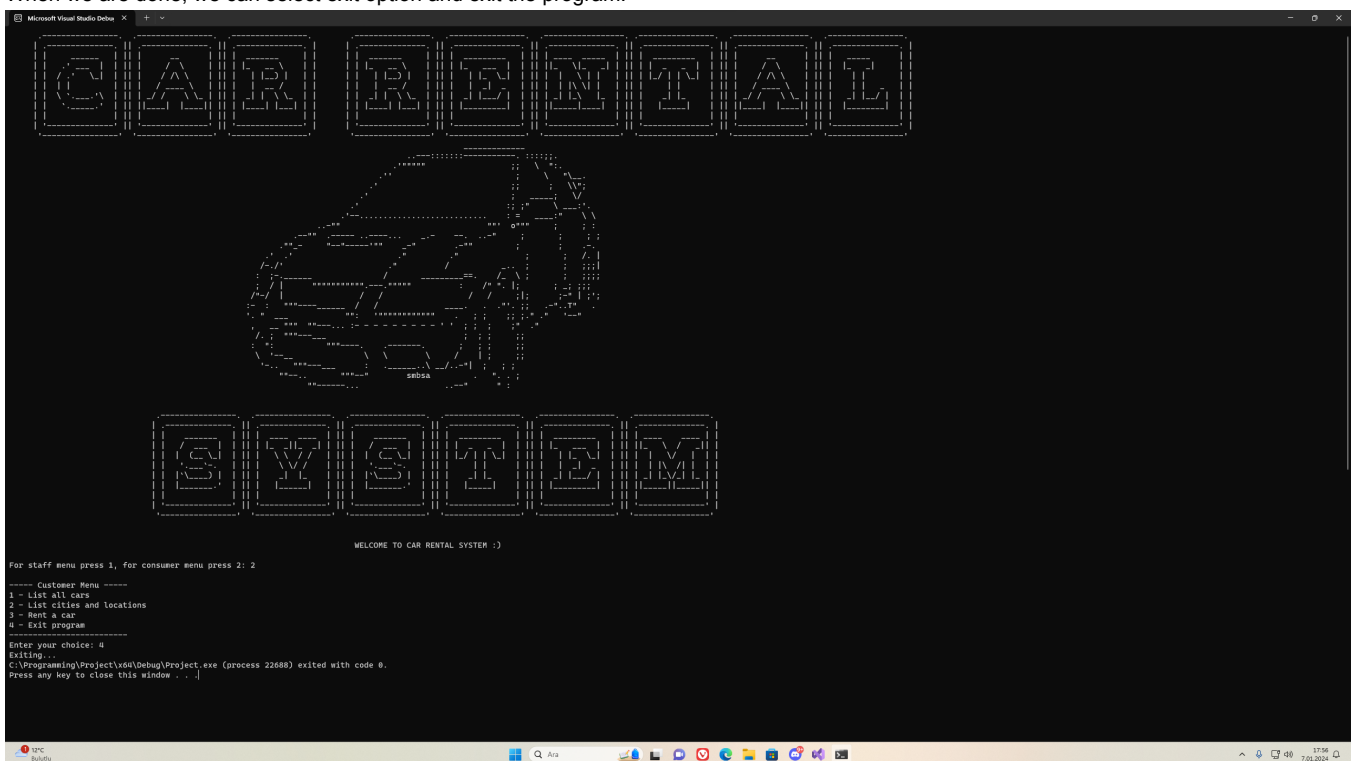
Customer name: Muhammed Selincan Bicer
Customer credit card number: 2566857954123508
Customer CVC: 254
Customer credit card expiration date: 28/01
Do you want to continue? (Yes: 1, No: 0)
For staff entry press '2': 2
Enter Username: Berfin Sahin
Enter Password: 12345
Login successful!
1' For add a car
2' for istatistics
3' for delete a car
4' for log out and exit
4' for return to the customer part: 2

The most preferred car is : Fiat
The most preferred segment is: Economy
The least preferred car is : Audi
The least preferred segment is : Comfort
1' For add a car
3' for delist a car
4' for return to the customer part: |
```

Figure 9. Staff, Operation 2

In statistics screen; we can see the most preferred car, most preferred segment, least preferred car and least preferred segment.

When we are done, we can select exit option and exit the program.



```
Microsoft Visual Studio Debu
CAR RENTAL
SYSTEM
WELCOME TO CAR RENTAL SYSTEM :)

For staff menu press 1, for consumer menu press 2: 2

----- Customer Menu -----
1 - List all cars
2 - List cities and locations
3 - Rent a car
4 - Exit program

Enter your choice: 4
Exiting...
C:\Programming\Project\src\Debug\Project.exe (process 22688) exited with code 0.
Press any key to close this window . . .
```

Figure 10. Exiting screen

2.3 System Design

In the system, there are eight classes: customer, car, city, date, personnel, place, price, and vehicle classes. The vehicle class is the parent of the car class, and the place class is the parent of the city class. There is an inheritance relationship between them. The system aims to protect itself by using private and protected attributes. With constructors and some functions, these private and protected attributes can be used if desired. The system also incorporates data encapsulation. Additionally, we used get and set functions to access variables within the classes.

The class diagram representation of the software is as follows:



Figure 11. Class Diagram

3. ERRORS AND SHORTCOMINGS

The aim of our Car Rental System project is to automate and simplify the car rental process, enhancing the experience for both businesses and customers. However, our project has certain areas that can be improved.

Our project aims to automate the car rental process and simplify traditional methods, providing an enhanced experience for both businesses and customers. It successfully achieves the fundamental goals of streamlining car rental operations and offering a user-friendly experience for both administrators and customers. However, to further enhance its effectiveness, we can address the following limitations.

The primary and notable limitation is the lack of a comprehensive database. Since the system does not store data in a text file, it lacks detailed analytic and reporting capabilities that could provide valuable insights into rental trends, popular car models, and customer preferences. Although the program stores data internally, the absence of data storage in a text file means that the data will be lost once the program is closed. Adding these features could improve the system's functionality and assist businesses in making informed decisions.

Additionally, the current system lacks a mechanism for gathering feedback from customers. Customer feedback is crucial for improving services and addressing potential issues. Integrating a feedback system would be beneficial in evaluating customer satisfaction and identifying areas for improvement.

Furthermore, while the system includes basic security features, it does not have advanced security measures that require continuous monitoring and updates for the protection of sensitive data.

By addressing these limitations, the Car Rental System can evolve into a more comprehensive and robust solution, meeting the dynamic needs of businesses and customers in the car rental industry.

4. CONCLUSION

This project serves as an example tailored to the needs of institutions and users for managing car rental services. It demonstrates the application of concepts learned in the CENG241 course, such as functions, class structures, constructors, inheritance, encapsulation, access specifiers etc. The system is implemented using C++, an object-oriented programming language, and can be further developed to cater to specific needs, setting a precedent for users in the field.